

Sanjida Yasmin

Data Engineer & Researcher & Analyst

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Work Experience

Data Engineer (Part-Time)

May 2022 – Present

JOYNEXT GmbH, Dresden, Germany

- Designed and orchestrated automated ETL pipelines using Apache Airflow and Python to synchronise critical bill-of-materials (BOM) data between SAP (SOAP web services) and MSSQL production databases.
- Developed an ETL workflow to migrate repository data from SVN to MSSQL, leveraging PostgreSQL as a middleware database for efficient data transformation.
- Built a parsing engine for XML and PAR files to extract test limit values; used stored procedures to supply correct parameters to production stations based on brand and SAP part number.
- Implemented automated workflows with Prefect to synchronise scripts from Git repositories and execute tasks on remote servers.
- Managed the daily generation and upload of critical configuration files to SAP servers to support continuous production operations.

Research Assistant – Database Interface Programming

Dec. 2021 – Mar. 2022

Helmholtz-Zentrum Dresden, Germany

- Developed an optimised Oracle PL/SQL interface with a focus on query performance, usability, and data security.
- Conducted cross-platform testing and debugging using web-based tools and C++/Qt applications to ensure API reliability.
- Managed source code repositories via Git/GitLab and documented all functions and processes to ensure long-term maintainability.

Junior Oracle Developer

Feb. 2019 – Sep. 2020

UOCBD, Chittagong, Bangladesh

- Developed full-stack web modules using Oracle APEX for the frontend and Oracle Database (PL/SQL) for backend processing.
- Collaborated with clients to gather requirements and translated them into technical specifications.

Oracle Developer

Nov. 2018 – Feb. 2019

R-creation, Chittagong, Bangladesh

- Built comprehensive full-stack applications using Oracle APEX for responsive UIs and advanced PL/SQL for robust backend logic.

Education

M.Sc. Distributed Systems Engineering

2020 – 2025

Technische Universität Dresden, Germany

- Master's Thesis (at JOYNEXT GmbH): Predictive Fault Detection and Maintenance Recommendation System Using PLC Messages and Operator Logs (Grade: 1.7).
- Developed a supervised multi-class classification system using multimodal data (PLC logs and manual text logs) to predict maintenance actions and reduce downtime.

- Implemented XGBoost and a novel hybrid BERT model to handle high cardinality (1,628 distinct classes) and class imbalance using Focal Loss.
- Result: The hybrid BERT model outperformed all baseline models, achieving a Precision@5 of 96.87 % and a Macro F1-Score of 0.77.

B.Sc. Computer Science & Engineering

2014 – 2018

International Islamic University Chittagong (IIUC), Bangladesh

Technical Skills

- **Programming:** Python (Proficient), SQL (Proficient), Oracle PL/SQL, C++, C, Bash, C#, Java
- **Machine Learning:** TensorFlow, PyTorch, PySpark, Scikit-learn, XGBoost, BERT
- **Databases:** Oracle (11g/19c), PostgreSQL, MS SQL Server, Oracle APEX (Full Stack), NoSQL
- **Data & DevOps:** Apache Airflow, ETL Pipelines, Docker, CI/CD, Kubernetes, Grafana, Linux, SAP Integration
- **Tools:** Git, GitLab, SVN, SSDT, SQL Developer, Qt Creator
- **Domain Knowledge:** OWASP Top 10, Penetration Testing, TARA, Automotive SPICE 3.1

Languages

- **Bengali** – Native
- **English** – C1 (Professional Working Proficiency)
- **German** – B2 (Currently in progress)